Irrational Numbers

An Irrational number is a number that cannot be written as a ratio of two integers.

The test:
Does the decimal terminate or recurr?

Rational number - the decimal either terminates (ends), or recurrs (repeats).

Irrational number - the decimal doesn't terminate (end) or recurr (repeat).

Exercise 6.6
Are the following numbers rational or irrational?

$\frac{5}{7}$ = 0.714285714285714285714285714285714285

The numbers 714285 recurr (repeat) thus the number is a rational number.

$\sqrt{3}$ = 1.732050807568877293527446341505872

There appears to be no recurring numbers, thus $\sqrt{3}$ is an irrational number.

1 Is 0.818181818181818181818181818181818181 (= $\frac{9}{11}$) a rational number or an irrational number?

2 Is $\sqrt{2}$ a rational number or an irrational number?
$\sqrt{2}$ = 1.414213562373095048801688724209698

3 Is π (pi) a rational number or an irrational number?
π = 3.141592653589793238462643383279502
(π is the ratio of the circumference of a circle to the diameter of the circle.)

4 Is $\sqrt{4}$ a rational number or an irrational number?
$\sqrt{4}$ = 2.0

5 Is 0.615384615384615384615384615384615384 (= $\frac{8}{13}$) a rational number or an irrational number?

6 Is e (Euler's Number) a rational number or an irrational number?
e = 2.718281828459045235360287471352662

7 Is φ (The Golden Ratio) a rational number or an irrational number?
φ = 1.618033988749894848204586834365638

8 Is 0.578947368421052631578947368421052631578947 (= $\frac{11}{19}$)
a rational number or an irrational number?